

Vitreaclear: Reducing Vision Disturbance from Floaters

Al Mo'ther Al-Maadi, Devin Brown, Rushav Dash, Jaxon Gordon, Jasmine Huang



Floaters **DISRUPT/BLOCK** vision, hindering everyday tasks like reading, driving, and working.

What are floaters?
Shadows or specks that drift across the visual field, caused by collagen clumps in the eye.

- Causes:**
- Aging
 - Eye injury
 - Eye surgery

- Current solutions:**
1. **Monitoring:** No intervention
 2. **Vitrectomy:** Highly invasive surgical procedure
 3. **Yttrium Aluminum Garnet (YAG) laser:** Break up floaters

60% of people aged 60+ have floaters [1]

3.5x higher risk for nearsighted individuals [2]

There is a need for an effective non-invasive way to treat floaters.

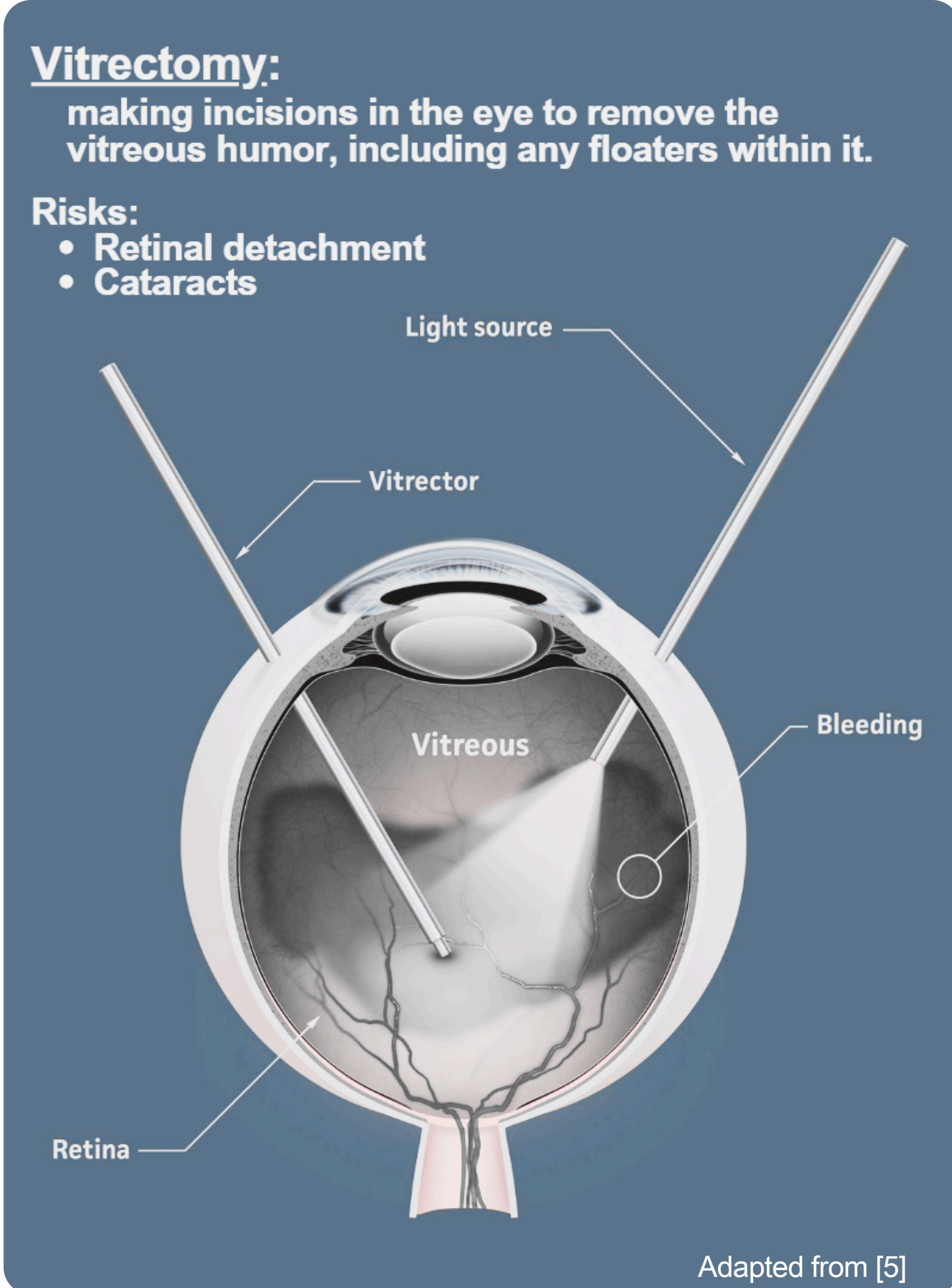
	Monitoring	Vitrectomy	YAG Laser	Our Solution
Reduces Floater Disturbance	✗	✓	⚠	✓
Non-invasive	✓	✗	✓	✓
No/Minimal Side Effects	✓	✗	✗	✓
Patient Comfort/Satisfaction	✗	✗	⚠	✓

Comparing Our Solution Against Current Floater Solutions



The **YAG laser** uses light energy to break up floaters.

- Limitations:**
- Unable to treat floaters near the eye's edges
 - Operator-dependent



Acknowledgments

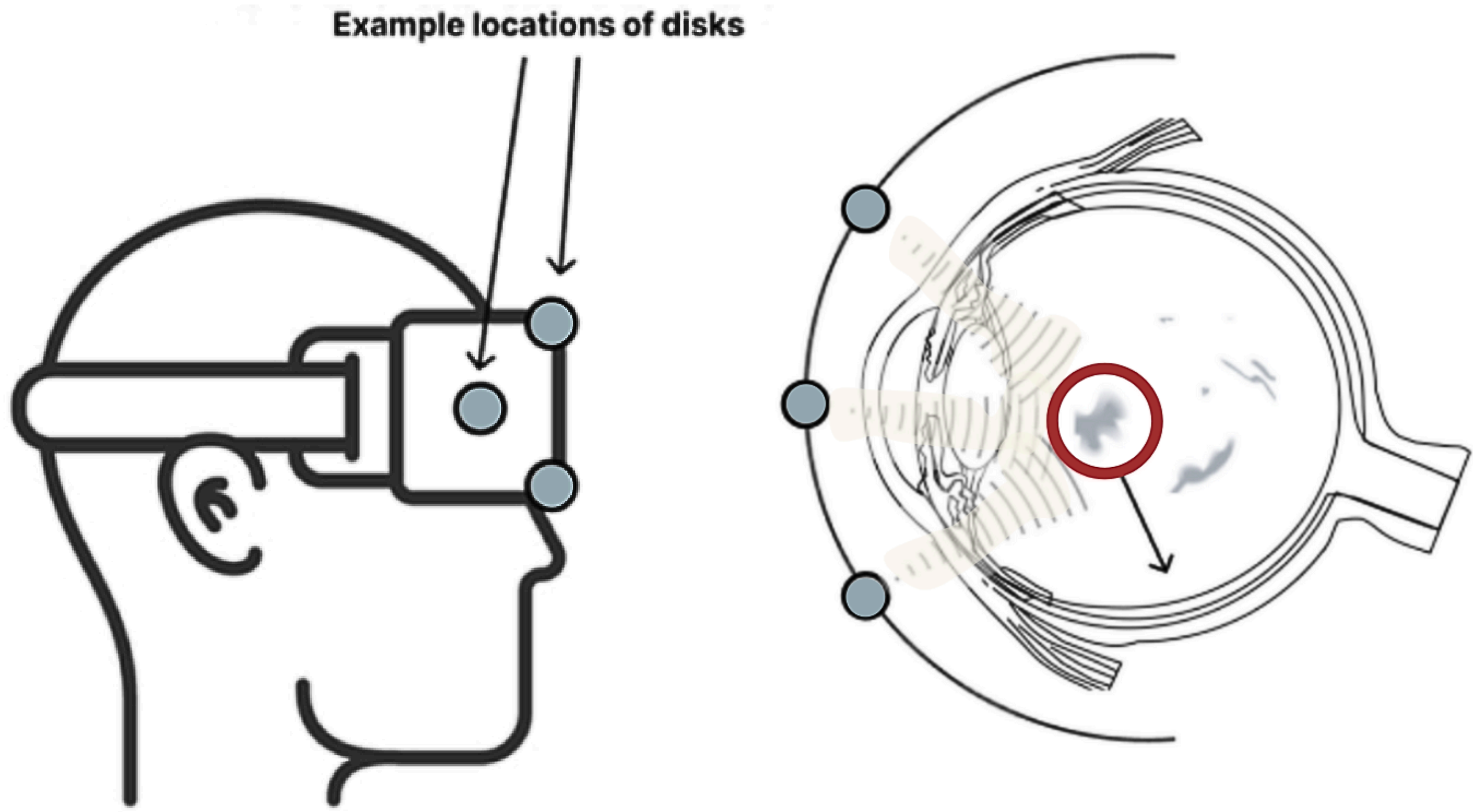
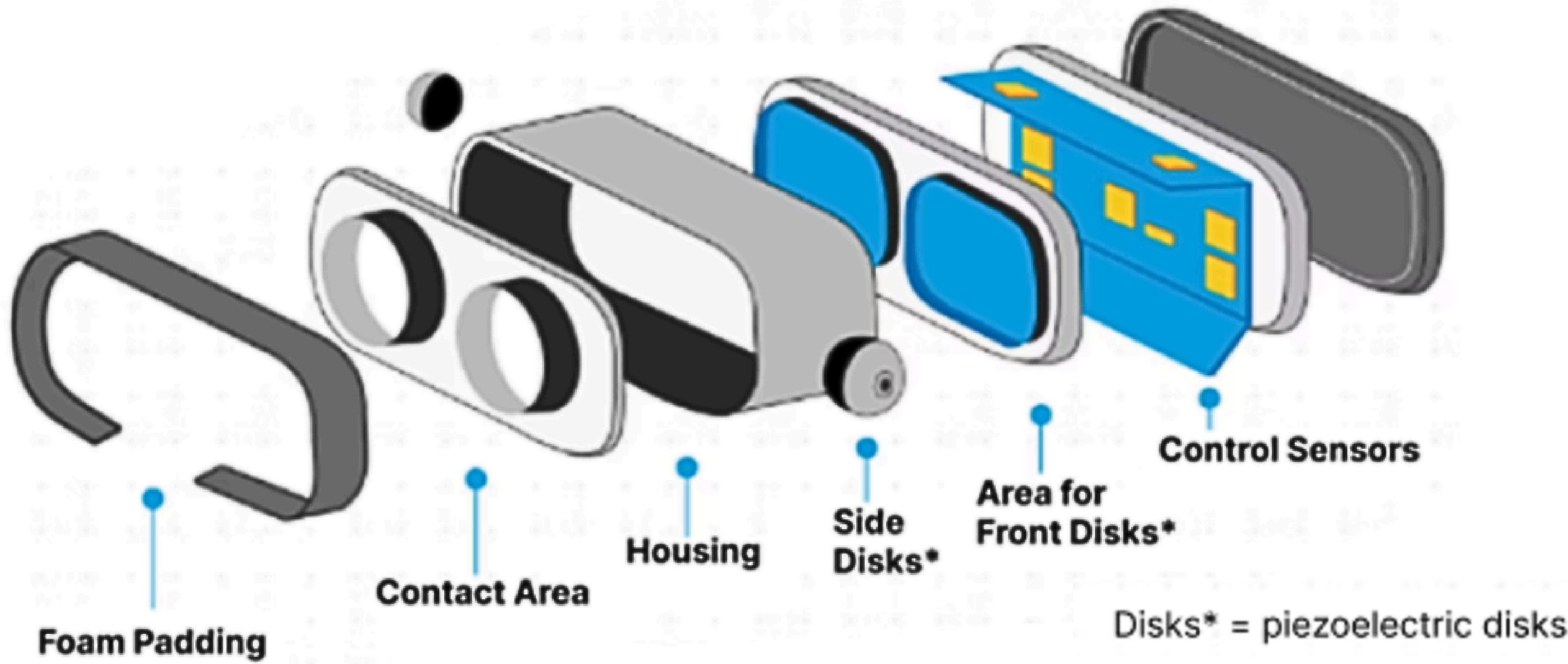
Advisors: Per Reinhall | Scott Lipsky | Shawn Swanson | Michael Malone | Eric Seibel
We would like to thank our advisors for their guidance and support.

References

- [1] D. Rodríguez, R. Canote, L. Izquierdo, R. Moncada, C. Maldonado, and A. Maria, "Quality of life results after vitreolysis in patients with symptomatic vitreous floaters," The Pan-American Journal of Ophthalmology, vol. 2, no. 1, p. 2, 2020, doi: https://doi.org/10.4103/pajo.pajo_24_19.
- [2] B. F. Webb, J. R. Webb, M. C. Schroeder, and C. S. North, "Prevalence of vitreous floaters in a community sample of smartphone users," International journal of ophthalmology, <https://pmc.ncbi.nlm.nih.gov/articles/PMC3693028/>.
- [3] C. C. medical professional, "Vitrectomy: Definition, Details & recovery," Cleveland Clinic, <https://my.clevelandclinic.org/health/treatments/24402-vitrectomy>.
- [4] "Vitrectomy - St. Adalbert's Hospital," Szpitalswwojciecha.pl, 2015. <https://www.szpitalswwojciecha.pl/en/zabieg/ophthalmology/vitreolysis/>.
- [5] "Guide to Vitrectomy: Costs, Timelines & More," NVISION, Mar. 21, 2019. <https://www.nvisioncenters.com/eye-surgery/vitrectomy/>.

Introducing VitreaClear: the non-invasive, sight-saving solution that applies acoustics in a novel way.

Vitreaclear's core ultrasound technology uses precisely arranged piezoelectric disks to generate controlled acoustic wave fields. These fields create pressure nodes that capture and reposition floaters to a non-disruptive area, restoring visual clarity without incisions.



Market Opportunity:

- TAM [\$33B]** – Total market spending (US) on floater treatments and consultations
- SAM [\$3B]** – Vitrectomy patient base (225,000/year) and their associated spending [3]
- SOM [\$72M]** – Attainable market based on initial outreach scope in WA state

Go-To-Market Strategy:

- **Regulatory Validation:** Pursue Class II 510(k) clearance.
- **Clinical & Professional Endorsement:** Establish credibility through professional network, publish in reviewed journals, present at conferences (AAO).
- **Adoption & Scaling:** Build clinical partnerships. Integrate into insurance reimbursement frameworks.